

Pratham Arora

[✉ pratham3992@gmail.com](mailto:pratham3992@gmail.com) [in LinkedIn](#) [GitHub](#) [Portfolio](#)

EDUCATION

Plaksha University

B.Tech. in Computer Science & AI — CGPA: 8.19 / 10.0

Mohali, India
Expected Aug. 2026

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, TypeScript, C#, HTML/CSS
Backend: FastAPI, Node.js, Next.js (SSR/CSR), REST, GraphQL
Databases: Azure Cosmos DB (NoSQL), PostgreSQL, Firebase, Vector DBs
DevOps & Tools: Git/GitHub, Linux, Azure Cloud, Vercel

EXPERIENCE

AI Intern (Backend Focus)

June 2025 – July 2025

Cotality | Kolkata

- Architected a scalable **FastAPI** backend to automate code documentation, handling AST analysis for 15+ languages.
- Designed optimized **NoSQL** document schemas in Azure Cosmos DB, selecting appropriate partition keys to balance read/write throughput and reduce Request Unit (RU) consumption.
- Designed RESTful API endpoints for the generative AI system, ensuring low-latency data retrieval for the frontend.

Software Development Engineer Intern

June 2024 – July 2024

Orangewood Labs | Noida

- Refactored the monolithic **RoboGPT** codebase into a modular architecture, reducing code dependency by 40% and improving maintainability for complex food preparation tasks.
- Integrated a Computer Vision stack with the control loop, optimizing calibration routines to significantly reduce robotic arm pose detection errors.

Freelance Web Developer

May 2025 – June 2025

Client Project | Remote

- Engineered a digital footprint on the **Vercel free tier**, implementing SEO best practices and reducing hosting costs to zero.
- Architected a modular **Firebase CMS**, empowering non-technical stakeholders to manage dynamic content without code intervention.

TECHNICAL PROJECTS

Resume Builder & Interview GPT | Next.js, TypeScript, Tailwind, Gemini API

Jan. 2025 – April 2025

- Developed a production-ready full-stack application featuring an AI-powered resume generator with real-time PDF preview.
- Implemented persistent user sessions using **Client-Side Storage (LocalStorage)** and secured authentication flows to maintain state across browser reloads without server overhead.
- Integrated Gemini API to provide "Interview GPT" functionality, offering personalized DSA guidance and feedback.

Reddit Data Ingestion Pipeline | Python, PRAW, Firebase, Data Engineering

Aug. 2025 – Dec. 2025

- Engineered a robust data ingestion pipeline using **PRAW** to scrape 22,000+ interactions from 19 Reddit communities.
- Designed a **NoSQL** schema in Firebase with real-time deduplication logic to maintain dataset integrity across concurrent scrapes.
- Implemented automated data cleaning and preprocessing scripts in Python, preparing unstructured text data for NLP analysis.

VR Conversational Agent | Unity, C#, Gemini API, Google Cloud STT/TTS, REST APIs

Aug. 2025 – Dec. 2025

- Engineered a real-time VR conversational system in **Unity (C#)** integrating cloud-based LLM inference with speech recognition and synthesis pipelines.
- Designed a low-latency interaction pipeline with asynchronous API orchestration and error handling for stable voice-based dialogue.
- Implemented embodied latency-masking mechanisms using procedural animations to maintain immersion during network and inference delays.
- Optimized end-to-end response time (avg. ~2s) through caching, parallel processing, and lightweight preprocessing.

LEADERSHIP

Head of Tech

April 2025 – Present

Athleda (Plaksha Sports Society) | Mohali

- Managed the full-stack development of the society website, overseeing code reviews, deployment pipelines, and architectural decisions.

Head of Web Development

Feb. 2024 – Feb. 2025

LEAP.AI@Plaksha | Mohali

- Led a team of developers in building and maintaining the AI club's web presence, mentoring juniors in **Next.js** and **Git** workflows.